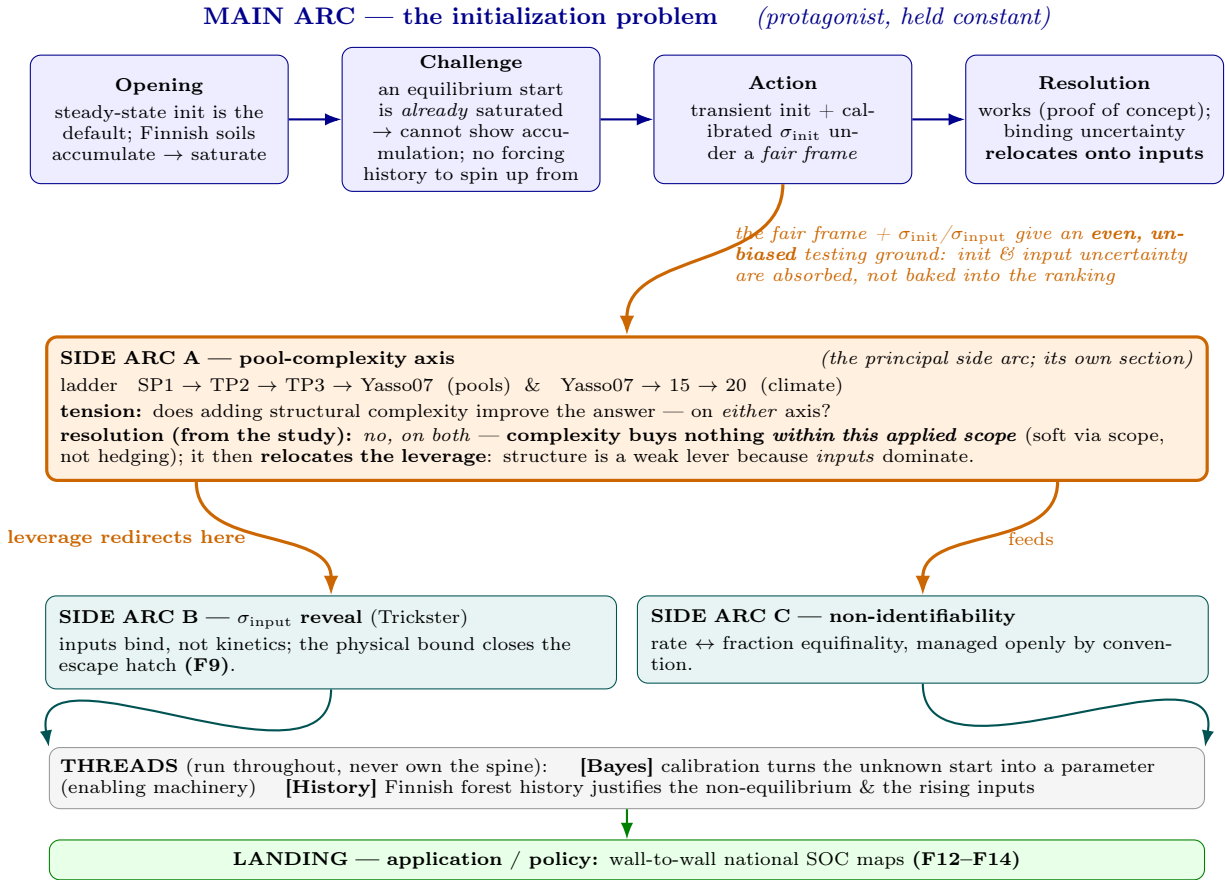


HIKET — figure storyboard (working draft)

Figures in narrative order (Schimel OCAR). Flux-bounded 20260710 run. Compiled July 14, 2026.

Protagonist = the initialization problem. *Trickster* = σ_{input} (input/history uncertainty). Very short notes per figure. [TO CREATE] marks gaps; **F9 is built**.

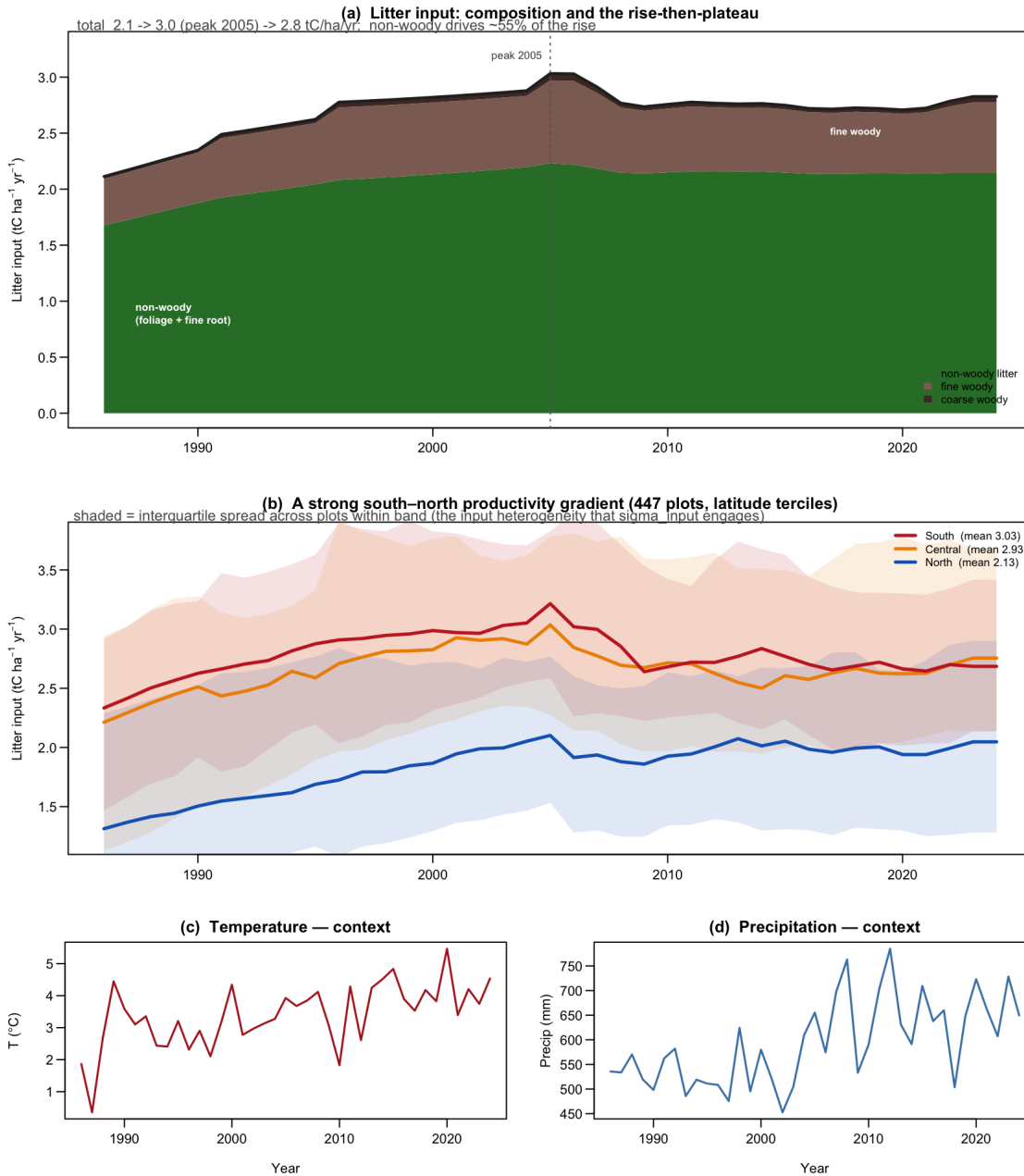
Storyline schema (main arc + side arcs)



Vertical link: the main arc's machinery (transient init + the two σ 's) is what makes SIDE ARC A a *fair* test — models compete on structure, not on who guessed inputs / initial state best.

Opening / Challenge — the phenomenon and the trap

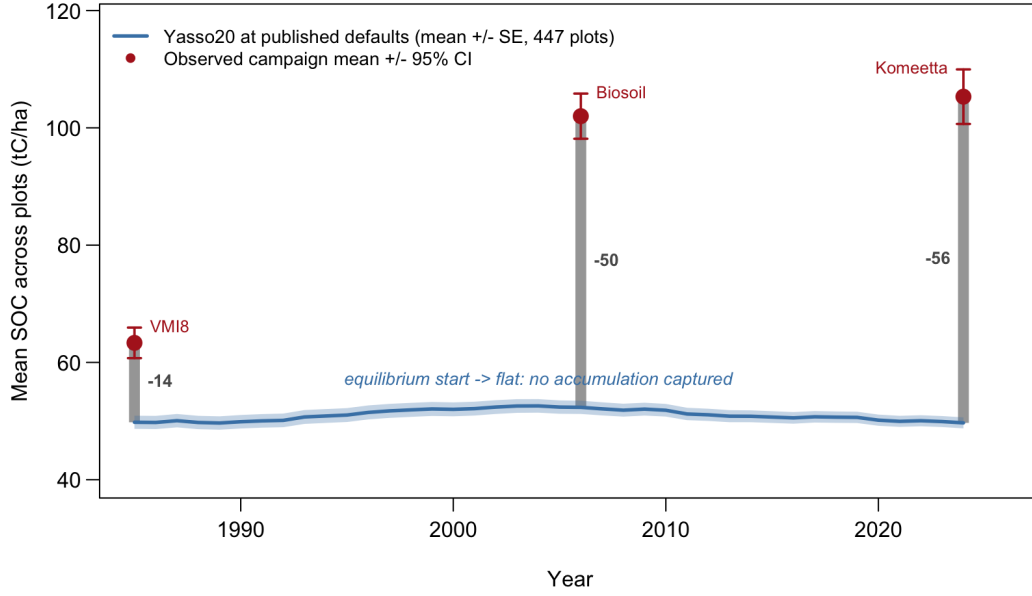
F1 · The inputs (redesigned — inputs-first).



What: three views of the litter input, with climate demoted to a supporting strip. **(a)** composition over time — stacked non-woody / fine-woody / coarse-woody: total rises 2.1 → 3.0 (peak 2005) then plateaus ~2.8; *non-woody foliage/fine-root litter drives ~55% of the rise*. **(b)** a strong south–north gradient (447 plots, latitude terciles: mean 3.0 / 2.9 / 2.1 tC/ha/yr) with the wide within-band plot spread shaded. **(c,d)** T and P — interannual swings, no accumulation trend — shown small, as context only. **Fits:** makes the opening about **inputs**, not climate. (a) the smooth rise-then-plateau ⇒ accumulation is input/history-driven (motivates transient init + the [History] thread); (b) inputs vary enormously in *space* — the heterogeneity that σ_{input} and site productivity (F11) engage, and the spatial face of the SOC maps (F12); (c,d) climate is jagged/trendless ⇒ any interannual SOC wiggle is climate, not accumulation. Foreshadows the σ_{input} reveal. *Script:* figures/build_F1_inputs.R (Yasso20 bundle — the size-class split lives only in the Yasso inputs).

F2 · Uncalibrated baseline under-predicts (redesigned — general, cross-plot mean).

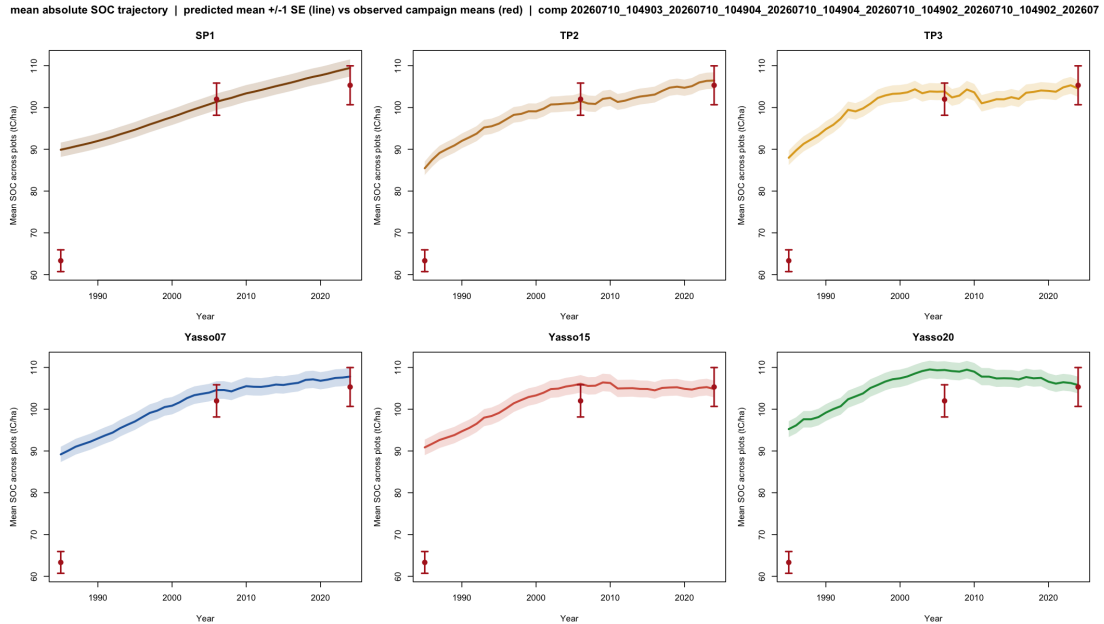
Uncalibrated baseline: published defaults miss the accumulation



What: Yasso20 forward at published defaults ($\sigma_{\text{input}} = 1$) with the conventional equilibrium start, *averaged over all 447 plots* (like F3, not 4 example plots). The baseline sits **flat at ~50 tC/ha**; the observed campaign means climb 63 → 102 → 105 — a widening deficit of −14/ −50/ −56 tC/ha. **Fits:** two failures in one glance. (1) **Flat** — an equilibrium start is *already* at its steady state, so it cannot accumulate: the OCAR *Challenge* made visual (motivates the transient-init protagonist). (2) **Low** — boreal-blind published defaults sit far below the stocks (the Mentor beat): the lineage is real but the *start*, not the soil, is wrong. *Script:* figures/build_F2_baseline.R (baseline predictions + observed campaign means; defaults are calibration-independent, so no RUN_ID coupling).

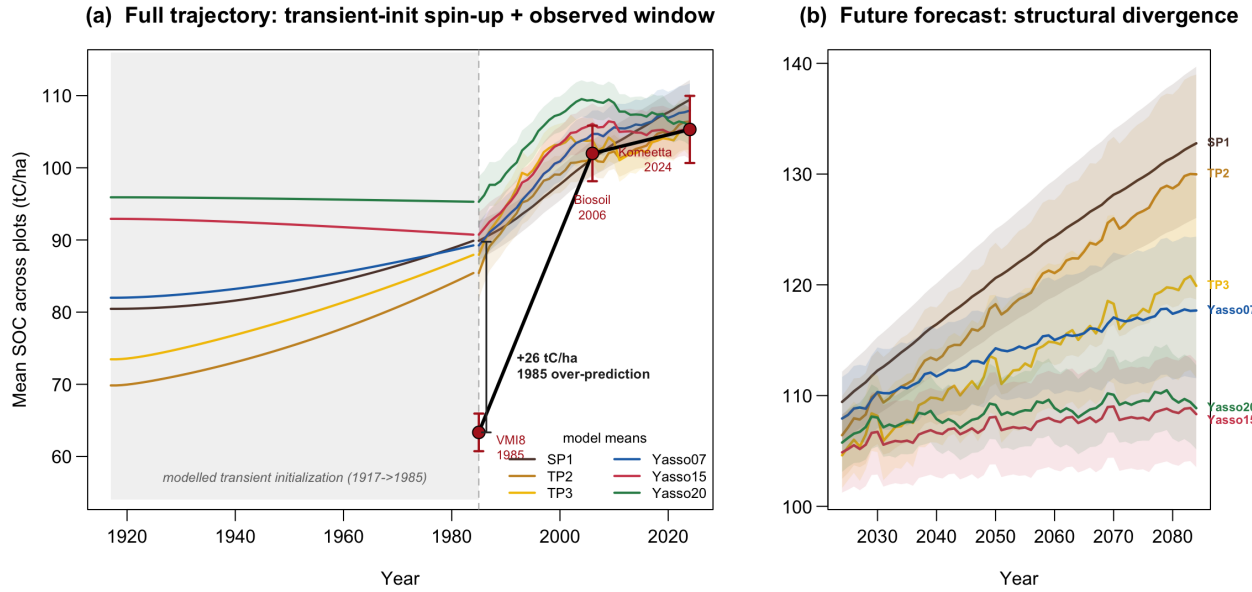
Action / Results — the protagonist pays off (H1)

F3 · Mean-SOC trajectories, six models (headline).



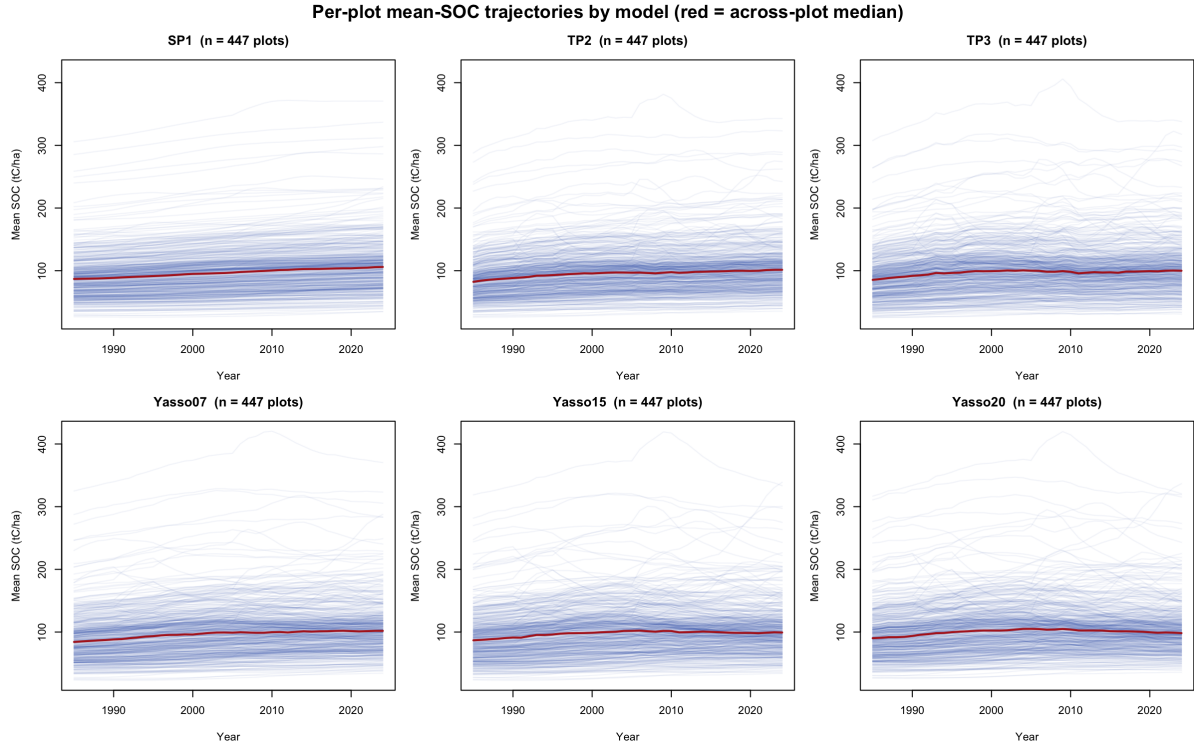
What: cross-plot mean SOC 1985 → 2024; all six rise then bend toward saturation, tracking campaign means. **Fits:** **H1** — transient init reproduces accumulation-then-saturation, the shape an equilibrium start forecloses. The central proof of concept.

F4 · The initialization problem (redesigned — 2-panel: past spin-up + future forecast).



What: six model means, same colour code, posterior ribbons in the observed/forecast windows. **(a)** the *full* trajectory including the modelled transient-init spin-up 1917 → 1985 (median line; constant climate + linearly ramped litter from a below-equilibrium 1917 anchor, level pinned to the calibrated 1985 stock) + observed window on the observed path (63 → 102 → 105): all models reach ~90 in 1985 — +26 tC/ha above the observed start — and climb only ~+0.70 tC/ha/yr through 2006 vs the data’s +1.84. **(b)** the *future forecast* 2025 → 2084: models agree in the calibrated window but *fan out sharply* (SP1 → 133 with no saturation; Yasso15/20 plateau ~108). **Fits:** the candid beat of the protagonist’s payoff. (a) makes the transient-init machinery visible: each model recovers from a below-equilibrium 1917 anchor — *but the structures disagree about how much* (simple models recover strongly from ~70–80; Yasso15/20 start near-full ~93–96), and all converge ~26 tC/ha above the observed 1985 start (the open [[transient-phase-open-question]], a live question not a failure). (b) is the structural-divergence story: equal where calibrated, divergent when extrapolated. *Secondary:* shape-divergence is expressiveness in the unobserved past/future, not data-constrained skill (honest lead-in to Side Arc A). *Reconciliation* of the earlier cumulative-average-rate figure: appendix “Reconciling the SOC-change diagnostics.” *Scripts:* build_F4_initialization.R (caches the spin-up reconstruction to F4_cache.rds), build_appendix_delta_reconciliation.R.

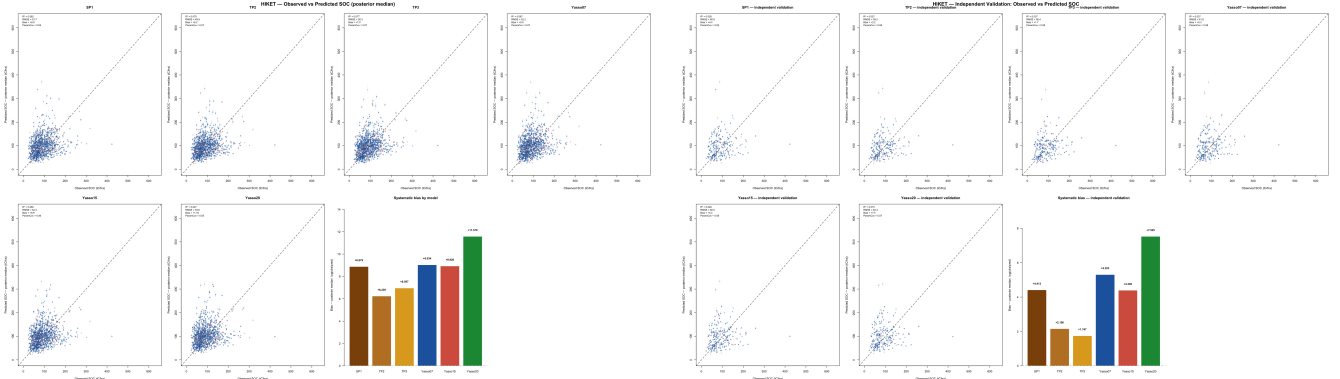
F5 · Per-plot SOC trajectories (supplementary / physical-sanity).



What: every plot's mean SOC; all models now peak at a physical ~ 100 tC/ha; TP2/TP3 jagged, SP1 smooth.
Fits: shows the flux bound worked (no inflated stocks) and that the TP3 oscillation is a real fast-pool signal, not the old Euler artefact.

Action / Results — structural skill under the fair frame (H2)

F6 · Observed vs predicted — calibration then holdout.



What: plot-level posterior medians vs observed; skill spread between models is small. **Fits:** **H2** — differences real but modest; on holdout the ranking nearly collapses (calib R^2 0.047–0.077; holdout 0.015–0.037). Bounding inputs cost skill *only* for TP2/TP3.

SIDE ARC A — Pool-complexity axis: does complexity improve the answer?

The principal side narrative — its own section. It carries a self-contained tension and resolution, and it serves the spine rather than competing with it.

Why the test is fair (link to the main arc). The comparison only means anything if the models are not secretly competing on who guessed the inputs or the initial state best. Transient init + the two auxiliary σ 's provide exactly that even ground: σ_{init} absorbs the unknown historical start, σ_{input} the litter-input uncertainty, both homogeneously and with uncertainty propagated. So the six structures “battle” on an unbiased field — the ranking reflects **structure**, not input/init luck.

The ladder — each rung a probe. **SP1 (1 pool)** the Occam floor: the minimal model already reproduces accumulation→saturation and is competitive; the study’s only identified rate — proof the *start*, not the structure, is the hero. **TP2 (2 pools)** the fast/slow split made visible: the pool-memory mechanism that justifies transient init; textbook-clean control. **TP3 (3-pool cascade)** the cautionary rung: the extra pool buys non-identifiability + input compensation + oscillation, not skill — the clearest escape-hatch victim. → **Yasso07/15/20** rates fixed; the operational destination (pools *and* the climate-integration sub-axis).

The tension. Does climbing structural complexity improve the answer — on *either* axis (pools SP1→Yasso07, climate Yasso07→15→20)?

The resolution — complexity buys nothing *within this scope* (soft but firm). *No, on both axes.* Skill does not climb either ladder: calibration R^2 stays 0.047–0.077 (holdout 0.015–0.037), the pool-axis OLS slope is *negative*, and the *most* complex model, Yasso20, is the *worst* fit on both calibration and holdout; the climate axis 07→15→20 falls monotonically 0.067→0.060→0.047. We do argue the anti-complexity claim — but the diplomacy is carried by **scope**, not by hedging.

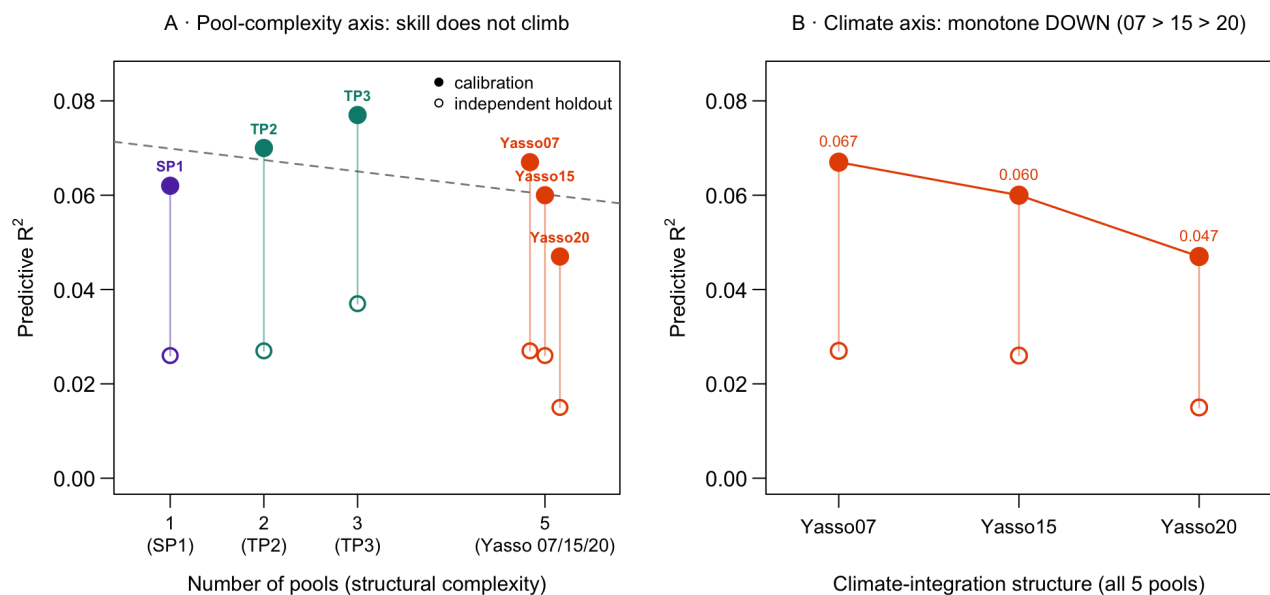
The scoped claim. For **this class of exercise** — an *applied* task whose aim is predictive accuracy *and* calibrated uncertainty on a defined, representative regional dataset, where site-specific **recalibration is affordable** — added structural complexity does not buy skill and can cost it. In that setting we *willingly* trade cross-region transferability for local fit: losing generalization power is an acceptable price when the target is one region done well. That is a claim about a *context*, not a universal verdict on model design.

Why this does not condemn Yasso (different objective function). The Yasso programme optimizes a *different* objective — generality, standardization, and transferability *without* local recalibration, plus litter-quality (AWEN) resolution — none of which this single-region SOC-fit test measures. Structural richness is exactly what buys *that*. So “more complexity does not help *here*” and “Yasso is the right inventory tool” coexist without contradiction; the two simply answer to different loss functions.

Why the claim is robust (three legs, not a bare correlation). (1) **Fair frame:** σ_{init} and σ_{input} absorb init/input uncertainty homogeneously, so it is genuinely *structure* being tested, not who guessed inputs best. (2) **Holdout:** the flat/declining ordering survives independent validation, not just in-sample. (3) **Mechanism:** the surplus degrees of freedom have a *known destination* — non-identified fractions (F7, F8), input compensation, and the TP3 oscillation (F9) — so we can say *why* complexity does not pay, not merely that it doesn’t.

The redirect. With that established, the arc does not crown a winner — it **relocates the leverage**: for this application the binding uncertainty is not in structure but in the *inputs and initial state*, which hands straight to Side Arc B (a relocated question, not a condemnation).

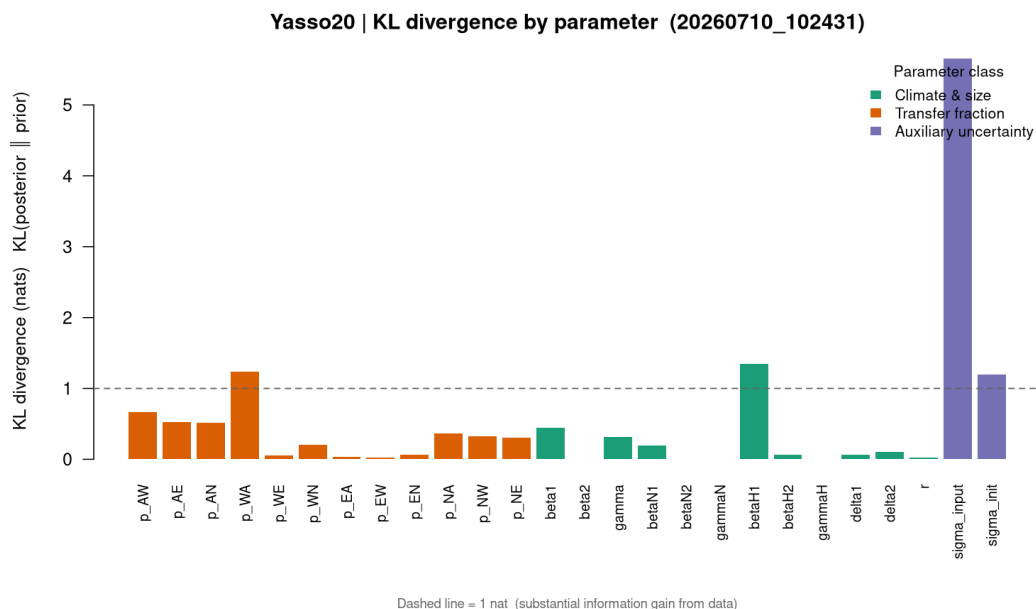
F6b · Skill vs. structural complexity. (NEW)



What: two panels, both R^2 (y): **A** pool count 1→5 (all six), **B** the climate sub-axis Yasso07→15→20. Filled = calibration, open = independent holdout. **Fits:** makes “complexity does not move the answer” visible in one glance — the resolution of SIDE ARC A. The pool-axis OLS slope is *negative* ($-0.0024 R^2/\text{pool}$); the climate axis falls monotonically $0.067 \rightarrow 0.060 \rightarrow 0.047$; the most complex model (Yasso20) is the worst on both calibration and holdout.

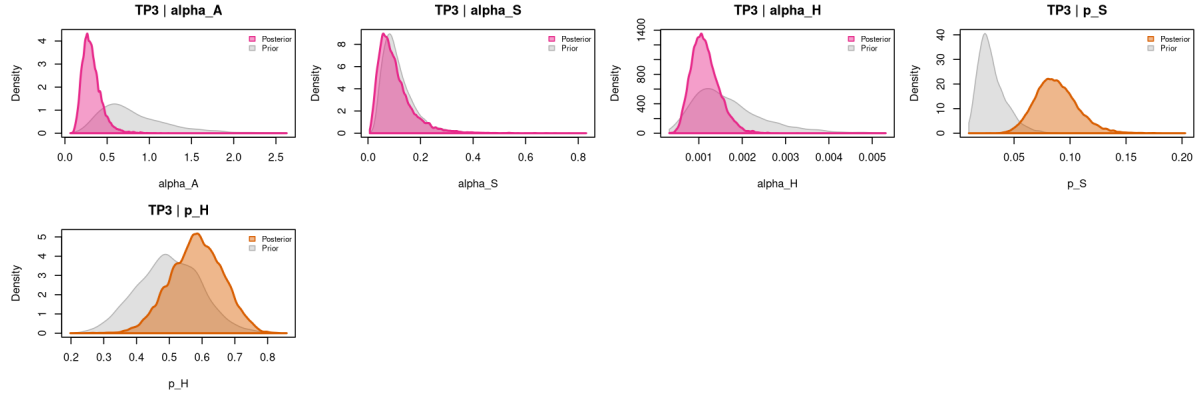
Action / Results — what the data can & cannot constrain (the hinge)

F7 · KL divergence — information gain per parameter.



What: posterior-vs-prior information gain by parameter; fractions gain ≈ 0 nats, σ_{input} gains the most. **Fits:** data constrains **inputs**, **not kinetics**. The identifiability lens; sets up the Trickster.

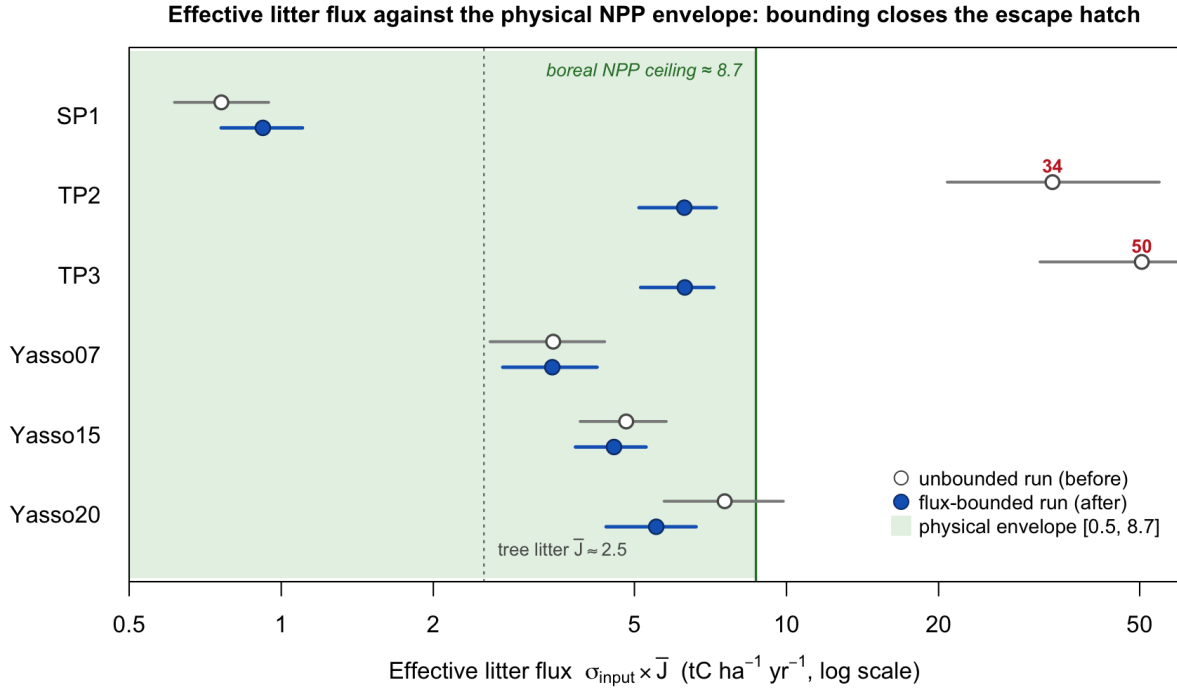
F8 · Fraction marginals — prior-pinning + ridges.



What: posterior $\approx$ prior for transfer fractions; near-perfect same-pool anti-correlations. **Fits:** the non-identifiability signature made visible — the Bayesian frame exposes it honestly instead of hiding it in a point estimate.

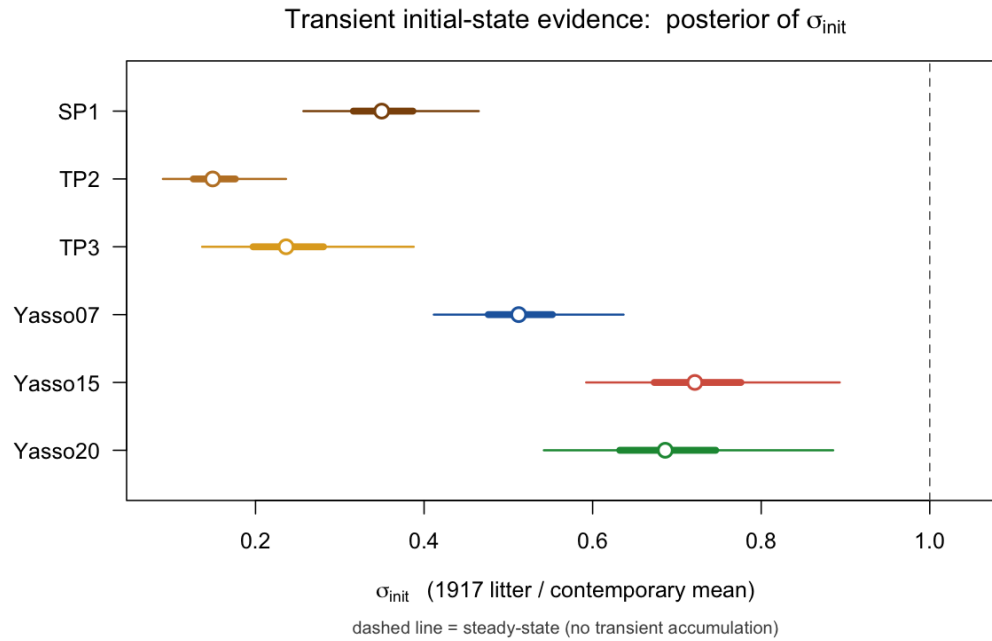
Resolution / Discussion — the σ_{input} reveal (Trickster payoff)

F9 · Effective litter flux vs the physical ceiling — before vs after. (NEW)



What: per model, effective flux $\sigma_{\text{input}} \times \bar{J}$ with the boreal NPP envelope [0.5, 8.7] shaded; unbounded (grey, before) vs flux-bounded (blue, after). **Fits: the quantitative payoff of the Discussion.** Unbounded TP2/TP3 hit 34/50 tC/ha/yr (impossible); the bound caps them at ~ 6.3 , and *only* their skill drops $\Rightarrow$ inputs, not kinetics, were doing the work.

F10 · σ_{init} caterpillar (historical anchor).



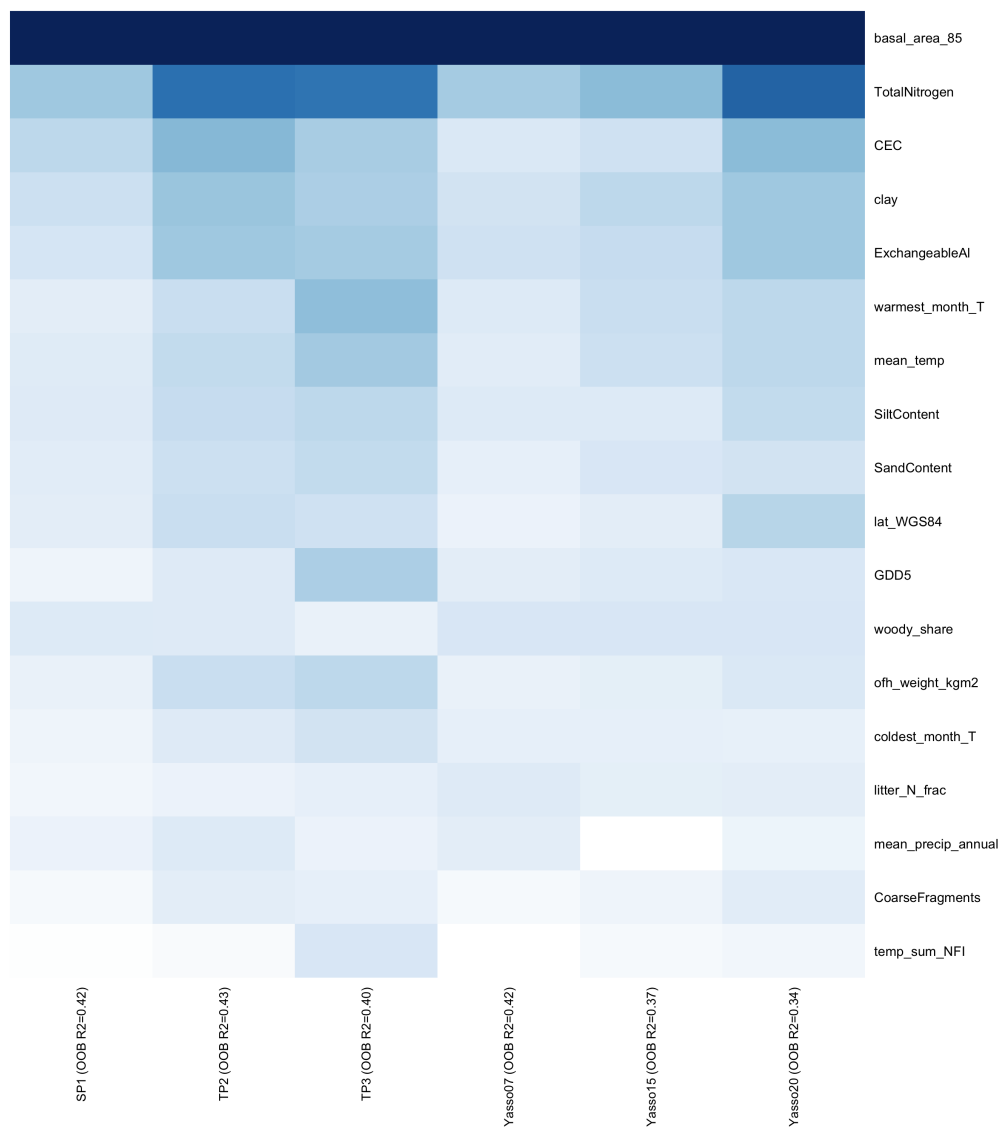
What: posterior σ_{init} per model; every 95% interval lies below 1. **Fits:** the *unrecoverable past* — all models infer a below-equilibrium 1917 start (post-exploitation recovery). Two-pool models lean hardest (0.15–0.25) vs Yasso (0.5–0.7).

F10b · [TO CREATE] Finnish forest history — post-war exploitation → managed growth ([History] thread; design TBD). **What:** a context/timeline figure — growing stock and management eras from the end of WWII to today. **Fits:** the *physical justification* for the non-equilibrium premise and the rising, non-stationary input history — why $\sigma_{\text{init}} < 1$ and why contemporary litter rises (F1). The [History] thread made concrete exactly where we discuss inputs.

Resolution — residual structure & where to go next

F11 · RF importance heatmap (residual drivers).

RF residual predictor importance (relative, within-model)



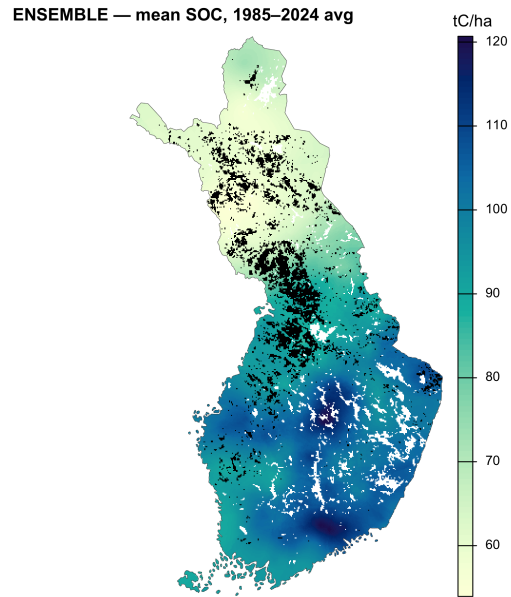
What: residual random-forest importances; stand basal area is the leading, diffuse, shared driver (5/6 models).

Fits: what's left is site/productivity structure — motivates stratifying the **forcing** (the optional strand), never the kinetics.

Application / Policy — wall-to-wall SOC (NextGenC)

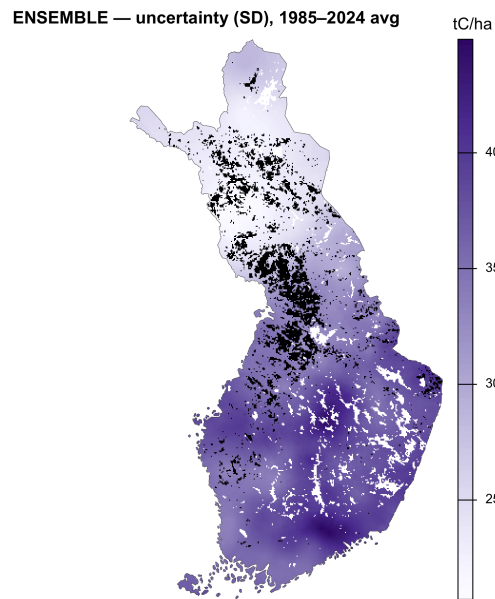
The policy-relevant deliverable: the operational product feeding the Finnish GHG inventory.

F12 · National wall-to-wall SOC — ensemble mean stock.



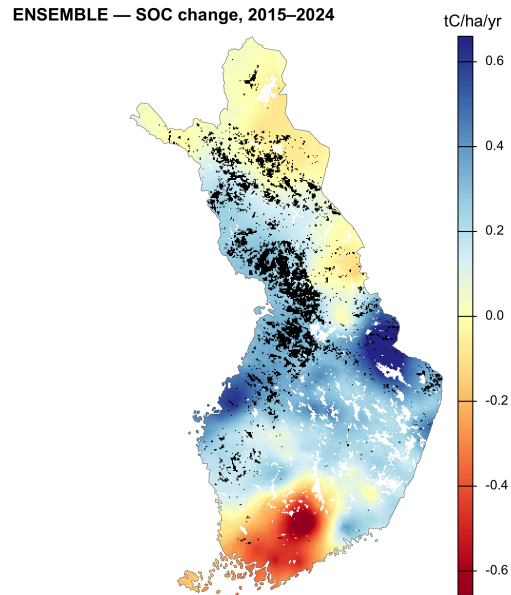
What: kriged 2 km ensemble-mean SOC over Finland; south-high/north-low gradient (black = peat/water masked).
Fits: the headline policy output — a wall-to-wall SOC stock estimate for the whole country.

F13 · Cross-model uncertainty (ensemble SD).



What: between-model spread per cell — where the six structures agree and where they don't. **Fits:** honest uncertainty mapped in space; the spatial face of “structure is a weak lever” (modest, structured spread).

F14 · SOC change — the “health” signal (ensemble 10-yr Δ).



What: wall-to-wall 10-year SOC change — gaining vs losing soils. **Fits:** the SOC-health map: whether the sink is accumulating or turning, spatially — the policy-facing payoff of F3 at national scale.

Decisions still open

1. **Side Arc A pitch: SETTLED.** We *do* argue “complexity buys nothing” — but **scope-bounded and soft:** the claim holds for an applied, recalibratable, single-region predictive task (accuracy + calibrated uncertainty), where trading cross-region transferability for local fit is acceptable. The Yasso programme answers a *different* loss function (generality/standardization/transferability without recalibration), which this test does not measure — so the two coexist. Robustness rests on three legs: fair frame, holdout, and a named mechanism for where the surplus DoF go.
2. σ_{input} **narrative inversion:** with the bound in Methods, “13–20× impossible” moves from a posterior *finding* to a *counterfactual* + the R^2 cost as live evidence. F9 carries it.
3. **F6b built** (figures/build_F6b_skill_vs_complexity.R). **F10b** (Finnish forest-history timeline, [History] thread) still to be designed — needs an external growing-stock / management-era source.

Revision notes (2026-07-14) — PENDING, not yet executed

User’s figure notes + agreed plan. Full detail in memory figure-revision-notes.

1. **F4 — forecast divergence = the dominant structural uncertainty (BIGGEST).** Simple models saturate less than Yasso; the *future* accumulation diverges hugely (SP1→133 vs Yasso15/20 ~108 by 2084). Politically central; un-adjudicable from current data. Does NOT contradict “complexity buys shape not skill” — it *sharpens* it: in-sample skill is blind to a structural choice that dominates the projection the inventory needs. New axis beyond skill = **extrapolation plausibility** (SP1’s unbounded rise is physically implausible; saturating structures more consistent with SOC saturation) — but Yasso’s plateau could be input-bound/humus-parking (most free DoF), not skill. Honest output = the **ensemble spread** (F12–14) as irreducible structural uncertainty. TODO: dedicated Discussion subsection; a *third beat* on the Side Arc A resolution; maybe elevate F4(b)/a projection figure to keystone; future-work note on testing extrapolation.
2. **F5 → supplementary** (sanity/robustness). Triage other figures for supp too.
3. **F6 → denser:** main text shows *holdout only* (calibration → supp); colour/class the points (latitude band / species / basal-area class) — ties to F11.

4. **F7** → **three Yasso panels** (07/15/20), text shrunk for readability; SP1/TP2/TP3 KL → appendix.
5. **F8** → **clarify + make readable**: state its role (non-identifiability signature = prior-pinning + same-pool ridges = Side Arc C; shows *where* surplus DoF go); fix legibility.
6. **F9** → **clarify shadings**: green band = physical NPP envelope [0.5, 8.7]; white = outside. Label the band on-figure; clarify before(grey)/after(blue) in caption.
7. **F11** → **central frontier arc**: basal area = #1 residual driver ⇒ residual structure points to site productivity / *inputs*, not kinetics; reinforces “inputs are the lever” (Side Arc B) + motivates stratifying the forcing (red strand). Discuss as “where to go next.”
8. **F12–13** → **merge** into one 2-panel figure (ensemble mean stock | ensemble SD).
9. **F14** → **maybe add a pool-recalcitrance / vulnerability panel** (discuss). “Health” via the pool distribution (labile/active vs stable share) = *intrinsic vulnerability*, a different axis from ΔSOC (current trajectory). Caveats: strongly model-structure-dependent (where the non-identified DoF live) so it carries large structural uncertainty; needs a common cross-model definition (active-pool share). Prototype + check spatial correlation with ΔSOC before committing (distinct signal ⇒ add panel; tracks ΔSOC ⇒ redundant).